CRITICAL ANALYSIS OF EXISTING WORK

Project Title

Machine Learning-Based Loan Risk Prediction: A Comparative Study of Ensemble, Oversampling, and Topological Approaches for Credit and Loan Default Assessment

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Machine Learning and Deep Learning have increasingly replaced traditional statistical approaches such as logistic regression and rule-based scorecards for assessing credit and loan default risk. While these traditional methods remain popular for their simplicity and interpretability, the reviewed literature shows that they struggle to capture non-linear relationships among borrower attributes and perform poorly on large, imbalanced financial datasets. Ensemble learning algorithms such as XGBoost, LightGBM, and Random Forest, along with deep learning architectures such as CNN, LSTM, ResNet, and DenseNet, have been widely adopted across personal credit, SME credit, maritime finance, and general banking domains, consistently outperforming conventional models in accuracy and discriminative power.

A recurring theme across the reviewed studies is class imbalance, since default cases form only a small fraction of most loan datasets. Different strategies have been proposed to address this: Zhuang and Wei (2024) used a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) to synthetically generate default samples before applying LightGBM, while Sayed et al. (2024) relied on SMOTE and SMOTE-TOMEK to balance the dataset before training ensemble and deep learning models. Yotsawat et al. (2021) took an alternative route by avoiding resampling altogether, instead assigning asymmetric misclassification costs within a Cost-Sensitive Neural Network Ensemble. Although each approach improved default detection to some degree, GAN-based oversampling is computationally expensive and risks generating unrealistic samples, SMOTE-based techniques can introduce noise near class boundaries, and cost-sensitive learning requires careful, dataset-specific tuning of cost parameters that does not always generalize well.

Beyond ensemble and deep learning approaches, a few studies explored more specialized directions. Zhao (2024) proposed a Consistent Assessment Method based on Linear Learning Analysis for micro and medium enterprise credit, which offered simplicity and interpretability but was limited by its linear formulation when applied to complex, non-linear financial patterns. Kheneifar and Amiri (2025) introduced Topological Data Analysis combined with Graph Neural Networks and classical classifiers for maritime loan default prediction, achieving a notably high ROC-AUC of 0.973 with XGBoost; however, this approach is domain-specific, computationally intensive, and was validated only on region-specific data that excluded broader economic and geopolitical variables.

A significant limitation across the reviewed literature is the limited integration of explainability. Only Kheneifar and Amiri (2025) applied SHapley Additive exPlanations (SHAP) to interpret feature contributions, while the remaining studies concentrated primarily on maximizing predictive accuracy without addressing how individual borrower or loan features drive the final decision. This lack of transparency is a critical gap for credit risk applications, where regulators, financial institutions, and borrowers require justifiable and auditable decisions rather than opaque black-box predictions.

Furthermore, most of the reviewed models were developed and validated on a single dataset or a narrow lending domain (personal credit, MME credit, or maritime finance), which raises concerns about their generalizability across diverse lending contexts. Deep learning models such as DenseNet and ResNet achieved the highest reported accuracy but demanded substantial computational resources, making them less practical for smaller financial institutions. Similarly, several authors explicitly acknowledged the need for testing on larger and more diverse datasets, as well as the inclusion of external factors such as economic and geopolitical conditions that are not captured by purely transactional or financial features.

In summary, the existing literature demonstrates that ensemble learning and deep learning techniques, combined with effective imbalance-handling strategies, can substantially improve loan risk prediction accuracy over traditional methods. However, the reviewed work reveals three persistent gaps: limited model interpretability due to minimal adoption of explainable AI techniques, restricted generalizability arising from single-domain or single-dataset validation, and a lack of comparative studies that evaluate ensemble, oversampling, and topological approaches together within a unified framework. These gaps motivate the proposed project, which aims to develop a comparative loan risk prediction system that combines robust ensemble and oversampling techniques with explainable AI, providing financial institutions with predictions that are both accurate and transparent across varied lending scenarios.